



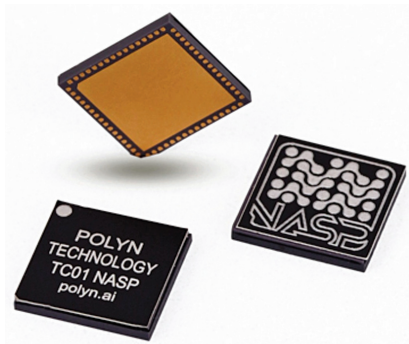
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

NASP chips

Always-on edge devices struggle with high power use and slow AI processing because conventional digital neural processors need complex computation



and energy-heavy conversions, which limit applications in wearables, robotics, automotive, industrial sensing, and audio devices. POLYN Technology addresses this with its NASP (neuromorphic analogue signal processing) technology, which runs trained neural networks directly in the analogue domain, processing sensor signals in their native form in microseconds at microwatt-level power while removing digital overhead. NASP chips are suitable for always-on edge devices and enable application-specific designs for audio, vibration, wearable, robotics, industrial, and automotive sensing, allowing for real-time AI inference with low power for efficient edge solutions.

Polyn Technology
<https://polyn.ai>

Super network interface card

AI servers often encounter various issues, including network congestion, limited bandwidth, multiple device hops, and GPU link failures. These challenges result in unused resources and increased costs. Traditional setups lead to inefficient traffic, poor load balancing, and network issues, thereby

increasing the total cost of ownership of AI clusters. Enfabrica has launched ACF-S super network interface card, addressing these problems. The



company claims to be the first to offer a 3.2 terabits per second multi-GPU solution, providing eight times the bandwidth for GPUs. In large clusters, it reduces device hops by up to 66%, improving operational efficiency while lowering capital expenditure by up to 29% and operating expenditure by as much as 55%, providing AI developers and data centre operators with a scalable, fault-tolerant GPU networking solution.

Enfabrica
<https://www.enfabrica.net>

Edge AI processors

Many connected devices face issues like limited computing power, slow data processing, high costs, and heavy reliance on cloud computing. This can



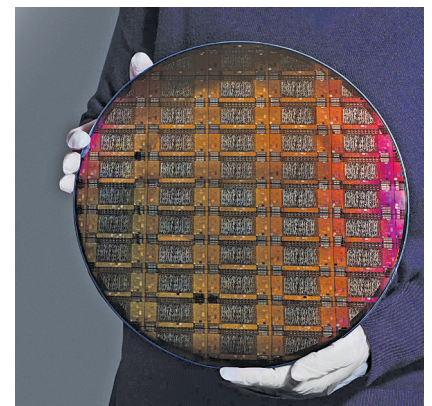
make running advanced AI locally difficult. The Astra SL2600 series of edge AI processors from Synaptics addresses this by enabling devices to process data

on-site, respond quickly, and operate efficiently with minimal reliance on the cloud. These processors support a wide range of applications, ranging from smart appliances, home and factory automation, autonomous robots, to charging infrastructure and healthcare devices. The company claims that the processors help developers and businesses to build smarter, faster, and more responsive devices at a lower cost.

Synaptics
<https://www.synaptics.com>

Quantum processors

IBM is addressing a key challenge in quantum computing with Quantum Nighthawk. It can handle circuits 30% more complex than its previous



system, with 120 qubits linked by 218 tunable couplers, supporting calculations with up to 5000 two-qubit gates while maintaining low error rates. Alongside this, IBM is testing Quantum Loom, an experimental processor that integrates all components needed for fault-tolerant quantum computing. Loom utilises classical hardware to decode errors in real time, as demonstrated at under 480 nanoseconds using qLDPC codes, enabling the scaling of high-fidelity superconducting qubits. These advancements aim to help researchers and organisations